

Janadri Yalla Yashwanth

Backend / Python Engineer

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SUMMARY

Backend engineer who ships production Python services end to end. At Marut Drones I build FastAPI systems backed by Celery, Redis, and PostgreSQL, stream large geospatial rasters from S3 with GDAL and rasterio, and serve computer-vision models behind async job APIs. I own features from API design through Docker packaging, CI/CD, and Linux deployment. B.Tech in CS (AI & ML), CGPA 8.6; available immediately.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript / TypeScript, C

Backend & APIs: FastAPI, Uvicorn (ASGI), REST APIs, async/await, Pydantic v2, SQLAlchemy 2.0

Async & Data: Celery, Redis, PostgreSQL / PostGIS, Alembic, S3 (AWS) / MinIO

Geospatial: GDAL, rasterio, GeoTIFF / COG, GeoJSON, CRS / projections

Auth & Security: JWT, RBAC, CORS, request validation

DevOps: Docker, Docker Compose, GitHub Actions, CI/CD, systemd, Linux

ML integration: PyTorch, OpenCV, SAM-2, DeepForest (served behind APIs)

Tools: Git & GitHub, Claude Code, VS Code, Postman, Docker, GitHub Actions, Linux

EXPERIENCE

Software Engineer Trainee — Marut Drones

Feb 2026 – Present

- Built three production geospatial detection pipelines integrating three model families — SAM-2, DeepForest, and GroundingDINO/LangSAM — behind a FastAPI backend, with async job APIs that enqueue Celery tasks on Redis and stream live progress and cancellation.
- Derived per-object stockpile volumes (m3) and material classification from nDSM height integration, backed by a 7-factor confidence rubric and Union-Find cross-tile NMS dedup at 0.3 IoU.
- Streamed orthomosaics block-by-block from S3 via GDAL /vsi3/ (512 MB cache, GDAL_NUM_THREADS=ALL_CPUS), eliminating full-raster memory buffering for multi-gigabyte COGs.
- Tuned S3 multipart transfers (64 MB chunks, 25 concurrent threads, pooled boto3 connections) and moved Celery workers to adaptive concurrency (nproc-2) for faster large-file I/O.
- Implemented owner-scoped RBAC and JWT auth with Pydantic v2 validation and Alembic migrations; packaged the PyTorch/SAM-2 stack into a Docker image shipped via GitHub Actions to systemd GPU/CPU workers.

Machine Learning Intern — Rooman Technologies

Jan 2025 – Mar 2025

- Developed and evaluated a sleep-disorder classification model on physiological and behavioral data, reaching 98% accuracy, and delivered it through a web app for clinicians.

SELECTED PROJECTS

Flight Log Analyzer — Drone Diagnostics Backend

FastAPI, Python, asyncio, ReportLab, Docker

- Built a FastAPI service that ingests raw drone flight logs and returns generated PDF diagnostics, with async uploads, 500 MB limits, extension allow-listing, and format auto-detection by magic-byte sniffing.
- Wrote a binary parser from scratch for the ArduPilot/DataFlash format using Python struct, decoding 50+ message types and reconstructing real wall-clock timestamps from GPS-week data.
- Serialized a headless .NET converter under Mono and Xvfb with an asyncio lock, per-job temp directories, and background cleanup, then rendered diagnostics with ReportLab and Matplotlib.
- Containerized the service with Docker and Docker Compose for reproducible deployment (~7,200 lines, solo-built).

Aerial Image Object Detection — Multi-Model Benchmark

Python, PyTorch, Hugging Face Transformers, SegFormer, OpenCV

- Benchmarked 4+ segmentation models (SegFormer-B5/B2, an aerial-domain model, SAM+CLIP) on identical drone imagery and selected SegFormer-B5 (ADE20K) for the cleanest detections, cutting false positives roughly 3x versus over-segmenting baselines.
- Built the inference pipeline in OpenCV/NumPy — sliding-window tiling on panoramas up to 14,400x7,200 px, HSV sky-skipping, mask-to-instance contour extraction, and per-class NMS — emitting annotated images plus structured JSON.

EDUCATION

Sri Venkateswara College of Engineering, Tirupati

Dec 2021 – May 2025

B.Tech in Computer Science (Artificial Intelligence & Machine Learning) — CGPA: 8.6

Narayana Junior College, Hyderabad

Jun 2019 – Mar 2021

Intermediate (MPC) — 96%

CERTIFICATIONS

- CS50's Introduction to Programming with Python — Harvard University Online
- Databases: Topics in SQL — Stanford Online